CSCI 680

# Affective Computing

William & Mary · Fall 2026

**Meetings:** Tuesdays & Thursdays, 11:00 a.m. – 12:20 p.m.

**Location:** John E. Boswell Hall 220

**Credits:** 3

**Format:** Graduate research seminar (paper presentation + discussion)


**Instructor:** Ashley (Ye) Gao, Assistant Professor of Computer Science

**Email:** ygao18@wm.edu

**Web:** https://lindagaw.github.io

**Office:** *ISC 4 Office 2381*


**Office Hour:** Tuesdays & Thursdays, 9:00 a.m. – 10:30 p.m.

## 1. Course Description

Affective computing is the study of systems that recognize, model, generate, and respond to human emotion. This seminar surveys the modern, machine-learning-centered version of that field: how emotion is represented in learned features, how it is recognized from speech and other signals under realistic conditions, how it is controlled in generative models, and how the resulting systems fail when they leave the lab.

The course is organized around close reading of one to two technical papers per meeting, drawn primarily from ICML, NeurIPS, ICLR, ACL/EMNLP, ICASSP, and Interspeech. Roughly the first third of the semester establishes shared foundations: emotion theory and its critiques, corpora and labels, and self-supervised speech representations. The remainder moves through four active research areas — emotion-aware audio-language models, expressive and controllable speech synthesis, robustness under distribution shift, and interpretability of affective representations. The semester ends with student project presentations.

This is a discussion course, not a lecture course. Students are expected to have read the assigned papers carefully before each meeting.

## 2. Prerequisites

Graduate standing in Computer Science, or permission of the instructor. Students should have:

- Working knowledge of machine learning at the level of CSCI 516 (or equivalent), including neural network training, optimization, and evaluation.
- Comfort with Python and at least one deep learning framework (PyTorch preferred).
- Enough mathematical maturity to follow derivations involving probability, linear algebra, and basic information theory.

Prior background in speech or signal processing is helpful but **not** required. Necessary audio concepts will be introduced as they arise.

## 3. Learning Outcomes

By the end of the semester, students should be able to:

1. Read a technical paper in affective computing critically — identifying its claim, its evidence, its baselines, and the gap between what it demonstrates and what it asserts.
2. Explain the major families of methods for emotion recognition and emotion-controllable generation, and articulate the tradeoffs among them.
3. Present a research paper to a technical audience and lead a substantive discussion of it.
4. Situate a specific research contribution within the broader literature.
5. Design, execute, and report on an original research project, including a defensible experimental protocol.
6. Reason about the validity and ethical stakes of inferring emotional state from behavioral signals.

## 4. Course Format and Expectations

Each meeting has a presenter (a student) and a discussion. The presenter gives roughly 35–40 minutes covering the problem, method, experiments, and weaknesses, then facilitates about 30 minutes of discussion. Every other student arrives having read the paper and submitted a reading response.

The instructor presents in Weeks 1–2 to model the format, and will contribute context throughout, but the seminar belongs to the students.

### Reading responses

Due by 9:00 a.m. on the day of each class for which you are not presenting. Roughly 250–400 words, covering:

- A one-paragraph summary of the paper's central claim.
- One substantive critique — a methodological gap, an unconvincing experiment, an unexamined assumption, or a claim the evidence does not support.
- One question you want the seminar to discuss.

Responses are graded credit/no-credit on evident engagement. Your two lowest scores are dropped, no questions asked.

## 5. Assessment

| Component | Weight |
|---|---:|
| Paper presentations (1–2 over the semester) | 25% |
| Reading responses | 15% |
| Seminar participation | 15% |
| Project proposal | 5% |
| Midterm progress report | 10% |
| Final project (report + presentation) | 30% |

### Final project

An original research project, done individually or in pairs. Acceptable project types include a novel method with empirical evaluation, a careful reproduction and stress-test of a published result, or a rigorous empirical study of an open question (for example, how a particular class of model degrades under a specific kind of shift). Pure literature surveys are not acceptable.

**Key dates:**

- Project proposal (2 pages) — due Friday, September 18
- Midterm progress report (4 pages) — due Friday, October 30
- In-class presentations — December 1 and December 3
- Final report (8 pages, conference format) — due Friday, December 11

## 6. Schedule

Dates follow the W&M Fall 2026 academic calendar. Classes begin August 26 and end December 4; the first meeting of this course is Thursday, August 27. Fall Break (October 8–11), Election Day (November 3), and Thanksgiving Break (November 25–29) remove three of our meetings. November 23–24 are university-designated remote instruction days, so our November 24 meeting will be held online.

The instructor reserves the right to adjust readings; the calendar dates are fixed.

### Unit I — Foundations: What Is Being Measured?

**1 · Thu Aug 27 — Course introduction**

*Scope of the field, seminar mechanics, how to read and present a technical paper. Presentation sign-ups. Skim: Picard, R. W. "Affective Computing." MIT Media Laboratory Perceptual Computing Section Technical Report 321, 1995.*

**2 · Tue Sep 1 — Can emotion be inferred from behavior?**

— Barrett, L. F., Adolphs, R., Marsella, S., Martinez, A. M., & Pollak, S. D. "Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movements." Psychological Science in the Public Interest, 20(1):1–68, 2019.

*Read the Executive Summary and Sections 1–2 closely; skim the rest.*

**3 · Thu Sep 3 — Corpora I: acted and elicited emotion**

— Busso, C., et al. "IEMOCAP: Interactive Emotional Dyadic Motion Capture Database." Language Resources and Evaluation, 42(4):335–359, 2008.

— Cao, H., et al. "CREMA-D: Crowd-Sourced Emotional Multimodal Actors Dataset." IEEE Transactions on Affective Computing, 5(4):377–390, 2014.

**4 · Tue Sep 8 — Corpora II: naturalistic and in-the-wild data**

— Lotfian, R., & Busso, C. "Building Naturalistic Emotionally Balanced Speech Corpus by Retrieving Emotional Speech From Existing Podcast Recordings." IEEE Transactions on Affective Computing, 10(4):471–483, 2019.

— Mollahosseini, A., Hasani, B., & Mahoor, M. H. "AffectNet: A Database for Facial Expression, Valence, and Arousal Computing in the Wild." IEEE Transactions on Affective Computing, 10(1):18–31, 2019.

### Unit II — Learned Speech Representations

**5 · Thu Sep 10 — Self-supervised speech representation learning**

— Baevski, A., Zhou, Y., Mohamed, A., & Auli, M. "wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations." NeurIPS, 2020.

**6 · Tue Sep 15 — Masked prediction and full-stack pretraining**

— Hsu, W.-N., et al. "HuBERT: Self-Supervised Speech Representation Learning by Masked Prediction of Hidden Units." IEEE/ACM TASLP, 29:3451–3460, 2021.

— Chen, S., et al. "WavLM: Large-Scale Self-Supervised Pre-Training for Full Stack Speech Processing." IEEE JSTSP, 2022.

**7 · Thu Sep 17 — What do these representations actually encode?**

— Yang, S.-w., et al. "SUPERB: Speech Processing Universal PERformance Benchmark." Interspeech, 2021.

> **Project proposal due Friday, September 18.**

**8 · Tue Sep 22 — Prompting and parameter-efficient adaptation**

— Lester, B., Al-Rfou, R., & Constant, N. "The Power of Scale for Parameter-Efficient Prompt Tuning." EMNLP, 2021.

— Shi, J., Zhang, Y., & Gao, Y. "Contrastive Learning for Speech Emotion Domain Generalization via Soft Prompt Tuning." Interspeech, 2025.

## Unit III — Audio-Language Models and Emotion

**9 · Thu Sep 24 — Audio-language models I**

— Deshmukh, S., Elizalde, B., Singh, R., & Wang, H. "Pengi: An Audio Language Model for Audio Tasks." NeurIPS, 2023.

— Tang, C., et al. "SALMONN: Towards Generic Hearing Abilities for Large Language Models." ICLR, 2024.

**10 · Tue Sep 29 — Audio-language models II**

— Chu, Y., et al. "Qwen-Audio: Advancing Universal Audio Understanding via Unified Large-Scale Audio-Language Models." arXiv:2311.07919, 2023.

— Gong, Y., Luo, H., Liu, A. H., Karlinsky, L., & Glass, J. "Listen, Think, and Understand." ICLR, 2024.

**11 · Thu Oct 1 — Making audio LLMs emotion-aware**

— Du, H., Lu, S., Zhou, G., & Gao, Y. "Emotion-Aware Audio Large Language Models with Dual Cross-Attention and Context-Aware Instruction Tuning." Interspeech, 2025.

**12 · Tue Oct 6 — Structured prompting and low-resource affect**

— Shi, J., Du, H., Hong, Y. A., & Gao, Y. "Plug-and-Play Emotion Graphs for Compositional Prompting in Zero-Shot Speech Emotion Recognition." IEEE ICASSP, 2026.

— Du, H., Shi, J., Myerston, J., Lu, S., Zhou, G., & Gao, Y. "Role-Guided Annotation and Prototype-Aligned Representation Learning for Historical Literature Sentiment Classification." Findings of EMNLP, 2025.

> **Thu Oct 8 — No class (Fall Break, Oct 8–11).**

**13 · Tue Oct 13 — Adapting at test time**

— Wang, D., Shelhamer, E., Liu, S., Olshausen, B., & Darrell, T. "Tent: Fully Test-Time Adaptation by Entropy Minimization." ICLR, 2021.

— Shi, J., Du, H., Hong, Y. A., & Gao, Y. "EMO-TTA: Improving Test-Time Adaptation of Audio-Language Models for Speech Emotion Recognition." IEEE ICASSP, 2026.

## Unit IV — Generating Emotional Speech

**14 · Thu Oct 15 — Neural audio codecs**

— Zeghidour, N., Luebs, A., Omran, A., Skoglund, J., & Tagliasacchi, M. "SoundStream: An End-to-End Neural Audio Codec." IEEE/ACM TASLP, 2021.

— Défossez, A., Copet, J., Synnaeve, G., & Adi, Y. "High Fidelity Neural Audio Compression" (EnCodec). arXiv:2210.13438, 2022.

**15 · Tue Oct 20 — Codecs that preserve affect**

— Shi, J., Du, H., Song, X., Hong, Y. A., Zhang, Y., & Gao, Y. "AffectCodec: Emotion-Preserving Neural Speech Codec for Expressive Speech Modeling." Findings of ACL, 2026.

— Borsos, Z., et al. "AudioLM: A Language Modeling Approach to Audio Generation." IEEE/ACM TASLP, 2023.

**16 · Thu Oct 22 — Zero-shot text-to-speech**

— Wang, C., et al. "Neural Codec Language Models Are Zero-Shot Text to Speech Synthesizers" (VALL-E). arXiv:2301.02111, 2023.

— Shen, K., et al. "NaturalSpeech 2: Latent Diffusion Models Are Natural and Zero-Shot Speech and Singing Synthesizers." ICLR, 2024.

**17 · Tue Oct 27 — Style and expressivity**

— Li, Y. A., Han, C., Raghavan, V. S., Mischler, G., & Mesgarani, N. "StyleTTS 2: Towards Human-Level Text-to-Speech through Style Diffusion and Adversarial Training with Large Speech Language Models." NeurIPS, 2023.

> **October 26 is the last day to withdraw. · Midterm progress report due Friday, October 30.**

**18 · Thu Oct 29 — Aligning generative models with preferences**

— Rafailov, R., Sharma, A., Mitchell, E., Manning, C. D., Ermon, S., & Finn, C. "Direct Preference Optimization: Your Language Model Is Secretly a Reward Model." NeurIPS, 2023.

— Wallace, B., et al. "Diffusion Model Alignment Using Direct Preference Optimization." CVPR, 2024.

> **Tue Nov 3 — No class (Election Day).**

**19 · Thu Nov 5 — Preference optimization for emotional TTS**

— Shi, J., Du, H., Song, X., Hong, Y. A., Zhang, Y., & Gao, Y. "Emo-BPO: Emotion Bidirectional Preference Optimization for Diffusion-based Emotional TTS." Interspeech, 2026.

— Shi, J., Du, H., He, Y., Hong, Y. A., & Gao, Y. "Emotion-Aligned Generation in Diffusion Text-to-Speech Models via Preference-Guided Optimization." IEEE ICASSP, 2026.

## Unit V — Interpretability and Robustness

**20 · Tue Nov 10 — Opening the black box: interpretable emotion control**

— Huben, R., Cunningham, H., Smith, L. R., Ewart, A., & Sharkey, L. "Sparse Autoencoders Find Highly Interpretable Features in Language Models." ICLR, 2024.

— Du, H., Shi, J., Lu, S., Zhou, G., & Gao, Y. "Sparse Autoencoders for Interpretable Emotion Control in Text-to-Speech." ICML, 2026.

**21 · Thu Nov 12 — Domain adaptation**

— Ganin, Y., et al. "Domain-Adversarial Training of Neural Networks." JMLR, 17(59):1–35, 2016.

— Tzeng, E., Hoffman, J., Saenko, K., & Darrell, T. "Adversarial Discriminative Domain Adaptation." CVPR, 2017.

**22 · Tue Nov 17 — Distribution shift in deployed affect systems**

— Lee, K., Lee, K., Lee, H., & Shin, J. "A Simple Unified Framework for Detecting Out-of-Distribution Samples and Adversarial Attacks." NeurIPS, 2018.

— Gao, Y., Baucom, B., Gordon, K., Rose, K., Wang, H., & Stankovic, J. "E-ADDA: Unsupervised Adversarial Domain Adaptation Enhanced by a New Mahalanobis Distance Loss." IEEE SmartComp, 2023.

**23 · Thu Nov 19 — Affect sensing in the wild: health deployments**

— Gao, Y., Salekin, A., Gordon, K., Rose, K., Wang, H., & Stankovic, J. "Emotion Recognition Robust to Indoor Environmental Distortions Using Out-of-Distribution Detection." ACM Transactions on Computing for Healthcare, 2021.

— Koltermann, K., Clapham, J., Blackwell, G., Jung, W., Burnet, E., Gao, Y., Shao, H., et al. "Gait-Guard: Turn-aware Freezing of Gait Detection for Non-intrusive Intervention Systems." IEEE/ACM CHASE, 2024.

**24 · Tue Nov 24 — Beyond speech: multimodal and physiological affect (remote instruction day)**

— Tsai, Y.-H. H., Bai, S., Liang, P. P., Kolter, J. Z., Morency, L.-P., & Salakhutdinov, R. "Multimodal Transformer for Unaligned Multimodal Language Sequences." ACL, 2019.

— Mersa, O., Jiang, M., Gao, Y., Li, Q., & Zhang, Y. "Hierarchical Convolution Multibranch Transformer for EEG Signals." IEEE ICASSP, 2026.

> **Thu Nov 26 — No class (Thanksgiving Break, Nov 25–29).**

### Unit VI — Project Presentations

**25 · Tue Dec 1 — Final project presentations, session I**

**26 · Thu Dec 3 — Final project presentations, session II**

> **Final report due Friday, December 11. No final exam. December 4 is the last day of classes; the university exam period runs December 7–11 and 14–15.**

## 7. Supplementary Reading

Optional; useful for project work and for filling gaps in background.

Zadeh, A., Chen, M., Poria, S., Cambria, E., & Morency, L.-P. "Tensor Fusion Network for Multimodal Sentiment Analysis." EMNLP, 2017, pp. 1103–1114.

Bagher Zadeh, A., Liang, P. P., Poria, S., Cambria, E., & Morency, L.-P. "Multimodal Language Analysis in the Wild: CMU-MOSEI Dataset and Interpretable Dynamic Fusion Graph." ACL, 2018, pp. 2236–2246.

Liang, S., Li, Y., & Srikant, R. "Enhancing the Reliability of Out-of-Distribution Image Detection in Neural Networks" (ODIN). ICLR, 2018.

Kong, Z., Goel, A., Badlani, R., Ping, W., Valle, R., & Catanzaro, B. "Audio Flamingo: A Novel Audio Language Model with Few-Shot Learning and Dialogue Abilities." ICML, 2024.

Macdonald, C., Chu, Z., Stankovic, J., Zhou, G., Shao, H., & Gao, Y. "MiddleGAN: Generating Inter-domain Images with GANs." Transactions on Machine Learning Research, 2024.

Nair, T., Nafee, M. W., Jiang, M., Gao, Y., Chen, H., & Zhang, Y. "Confidence-Aware Ranker Ensembles for Robust In-Context Knowledge Editing." Findings of ACL, 2026.

Clapham, J., Zhou, M., MacDonald, C., Koltermann, K., Gao, Y., & Shao, H. "ElectroMeter: The Practical Electrolyte Measurement System." IEEE/ACM CHASE, 2025.

Ko, E., Gao, Y., Wang, P., Wijayasingha, L., Wright, K. D., Gordon, K. C., Wang, H., Stankovic, J. A., & Rose, K. M. "Challenges to Recruiting Dementia Caregiving Dyads in Community-Based Settings." JMIR Formative Research, 2025.

## 8. Policies

### Attendance

This is a seminar; attendance is the course. More than two unexcused absences will materially affect your participation grade. If you must miss a meeting, notify the instructor in advance. Absences for illness, religious observance, or documented emergency are excused.

### Late work

Reading responses are not accepted late — the point is that you have read the paper before we discuss it. Two dropped scores exist to absorb ordinary life. Project deliverables may be submitted up to 48 hours late with a 10% penalty; beyond that, only by prior arrangement.

### Generative AI

You may use AI tools for coding assistance, debugging, and language polishing on written work. You may **not** use them to generate your reading responses, your critique of a paper, or the substantive content of your project report — those are the thinking the course is designed to teach. Any substantial AI assistance must be disclosed in a brief note at the end of the submission. Undisclosed use is an honor code matter.

Note also that AI tools fabricate citations with some regularity. You are responsible for every reference you submit.

### Honor Code

All work is governed by the William & Mary Honor Code. Collaboration on project work is encouraged within your declared team; representing others' work or ideas as your own is not permitted. When in doubt about whether a form of collaboration is acceptable, ask before rather than after.

### Accessibility

William & Mary accommodates students with disabilities. Students seeking accommodations should contact Student Accessibility Services to establish eligibility, then share the accommodation letter with the instructor as early in the semester as possible so arrangements can be made in time to be useful.

### Wellbeing

Graduate study is demanding, and a seminar that meets twice weekly makes it hard to disappear quietly during a difficult stretch. If something is going on, tell me early — nearly every problem is easier to solve in September than in November. The W&M Counseling Center is available to students at no cost.

### Research ethics

Several projects in this course will involve human affective data. Any project collecting new data from human participants requires IRB review before collection begins; plan for this timeline in your proposal. Projects using existing corpora must respect the license and use restrictions of those corpora, several of which restrict redistribution and commercial use.

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*Syllabus subject to change. Any changes to readings or deadlines will be announced in class and posted to the course site.*